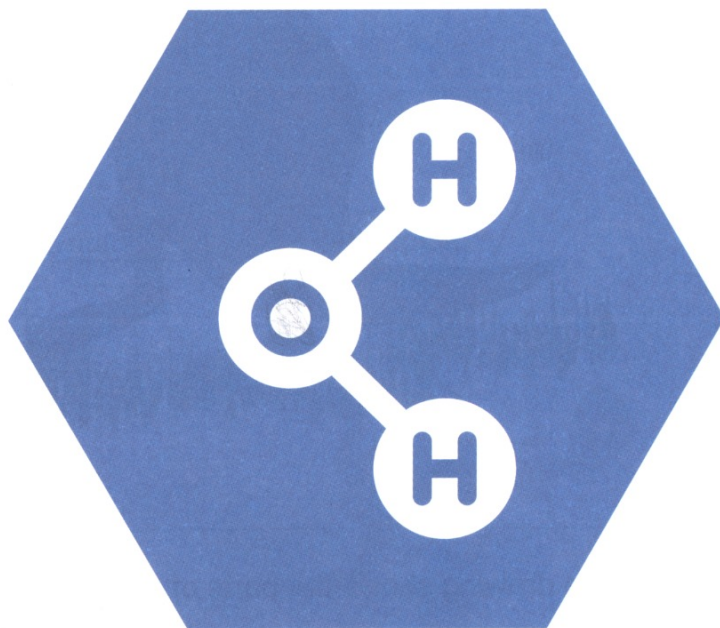




UNSW Global
AUSTRALIA

INTRODUCTORY PAPER



2016
ICAS

International Competitions
and Assessments for Schools

SCIENCE

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

30 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.

Use a 2B or B pencil.

Do **NOT** use a pen.

Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.

Your score will be the number of correct answers.

Marks are **NOT** deducted for incorrect answers.

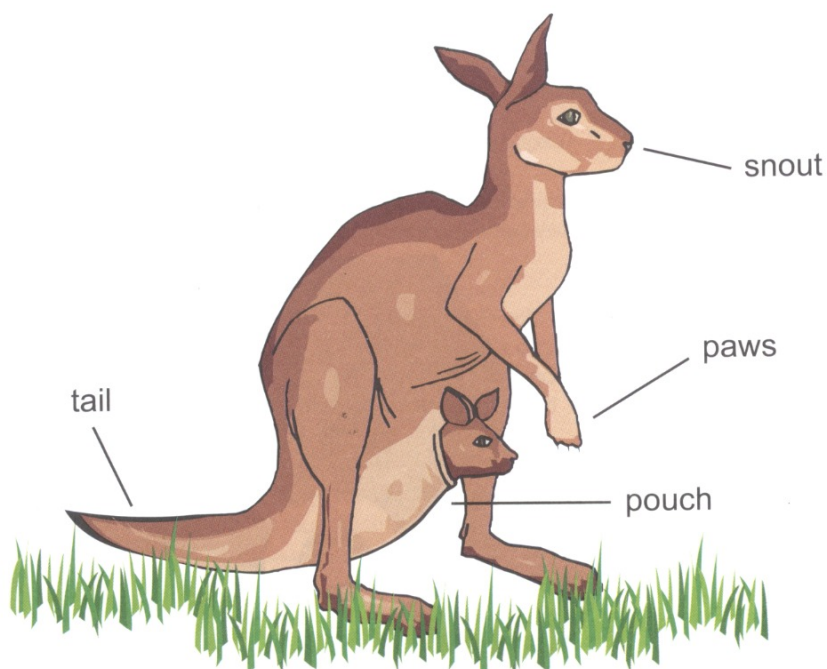
There are **30 MULTIPLE-CHOICE QUESTIONS** (1–30).

Use the information provided to choose the **BEST** answer from the four possible options.

On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a calculator and a ruler.

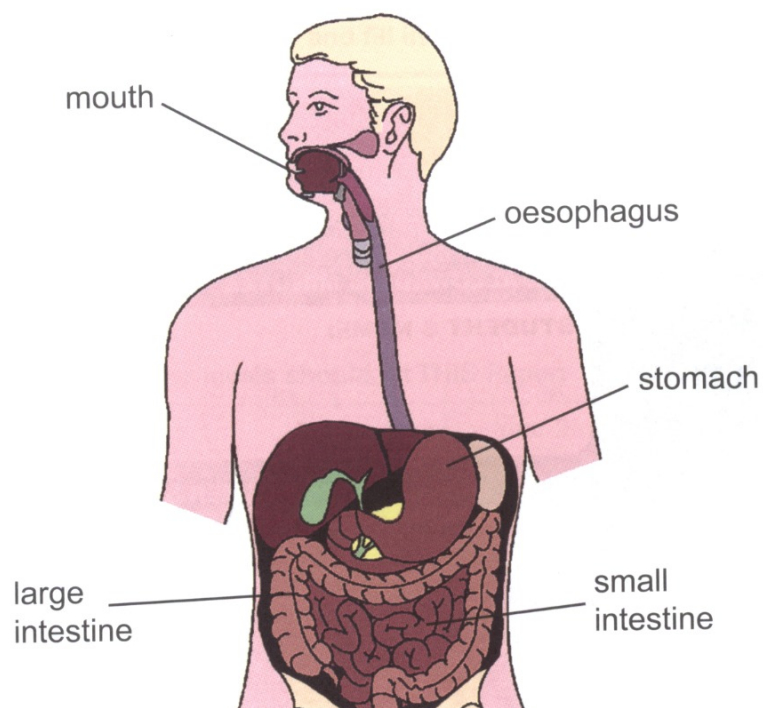
1. The drawing shows a kangaroo.



Which part of a kangaroo helps support its body while it is standing up?

- (A) snout
- (B) paws
- (C) pouch
- (D) tail

2. The drawing shows the parts of our body that help break down the food we eat.



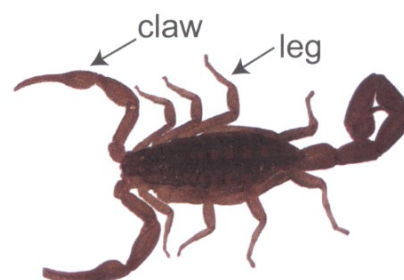
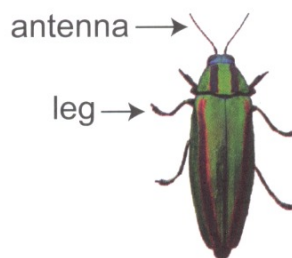
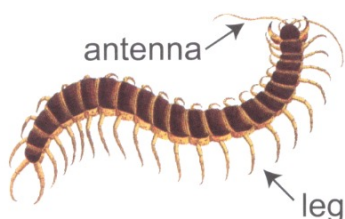
Where does food start to be broken down?

- (A) mouth
- (B) oesophagus
- (C) stomach
- (D) small intestine

3. This is a spider.

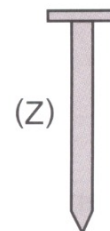
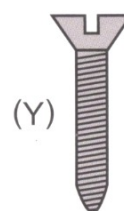
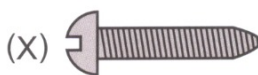
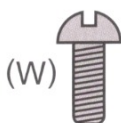
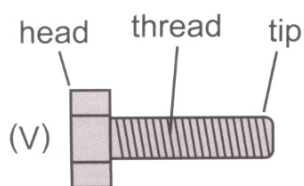


How many of the animals below have the same number of legs as the spider?



- (A) 0 (B) 1 (C) 2 (D) 3

4. Lily had these objects in her toolbox. One object does **NOT** belong with this group.



Which object does **NOT** belong to this group and why?

	Object	Why it does NOT belong to the group
(A)	V	It has a flat head.
(B)	Z	It doesn't have a flat tip.
(C)	W	It has a round head.
(D)	Z	It doesn't have a thread.

5. Some animals, such as bears, hibernate over winter. This means that they go into a sleep and do not hunt for months.



What must bears do before hibernating over winter?

- (A) store food to eat while they sleep
- (B) eat a lot to store fat in their body
- (C) dig a very deep burrow underground
- (D) exercise to make up for sleeping for a long time

6. When air is cooled down, it is easier to squeeze it into a smaller space.

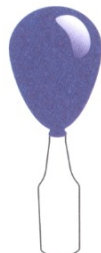
Mr Boot blew up a balloon and fitted it over the top of an empty bottle, as shown.



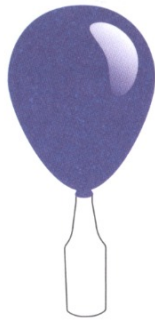
He then put the bottle in some icy water.

What did the balloon look like after the bottle had been in icy water?

- (A)



(B)



(C)



(D)

Pop!



7. Diana wants to know the kind of pet people in her school like most.

She does not have time to ask everybody in her school.

Which of these would be the best way to find out?

- (A) Ask her friends what their favourite kind of pet is.
- (B) Ask only girls what their favourite kind of pet is.
- (C) Ask only younger students what their favourite kind of pet is.
- (D) Ask some boys and girls from each year group what their favourite kind of pet is.

8. Sunil's teacher gave him six photos, each showing a different animal.



He told Sunil to think of a way to place the six animals into two groups.

Which of the following can he use to group these animals?

	Group 1	Group 2
(A)	with wings	without wings
(B)	with eyes	without eyes
(C)	big	small
(D)	old	young

9. Avril had four identical cans, each with a different-sized hole, as shown.

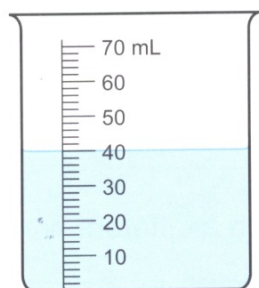
She poured an equal amount of water into each can. She then turned the cans upside down and timed how long each one took to empty.



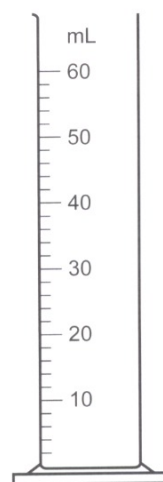
Which can took the longest time to empty?

- (A) can 1
- (B) can 2
- (C) can 3
- (D) can 4

10. Joe had two containers, X and Y.



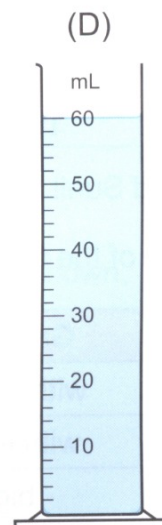
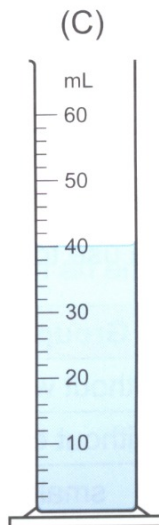
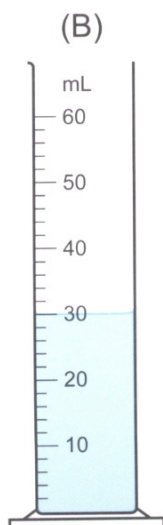
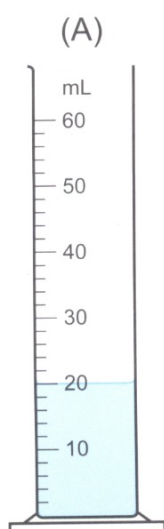
container X



container Y

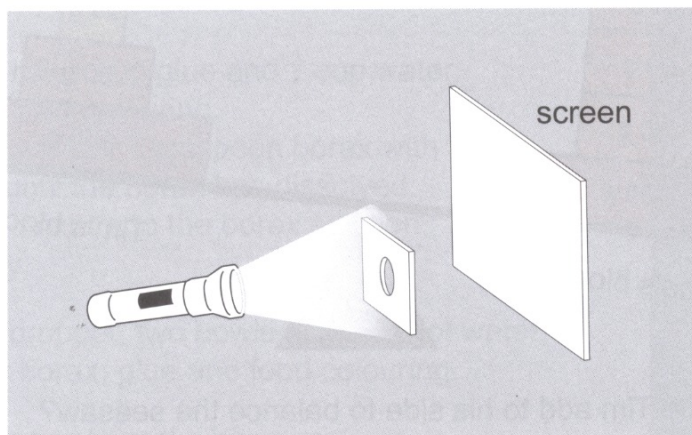
All the water from container X was poured into container Y.

Which diagram shows the water level in container Y?



11. Tina used her torch to make shadows on a screen. She had a piece of cardboard with a hole cut out of the middle.

Tina shone the light from the torch onto the cardboard as shown in the diagram.

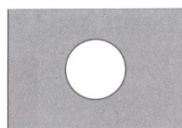


What did the shadow on the screen look like?

(A)



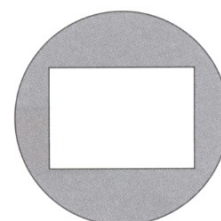
(B)



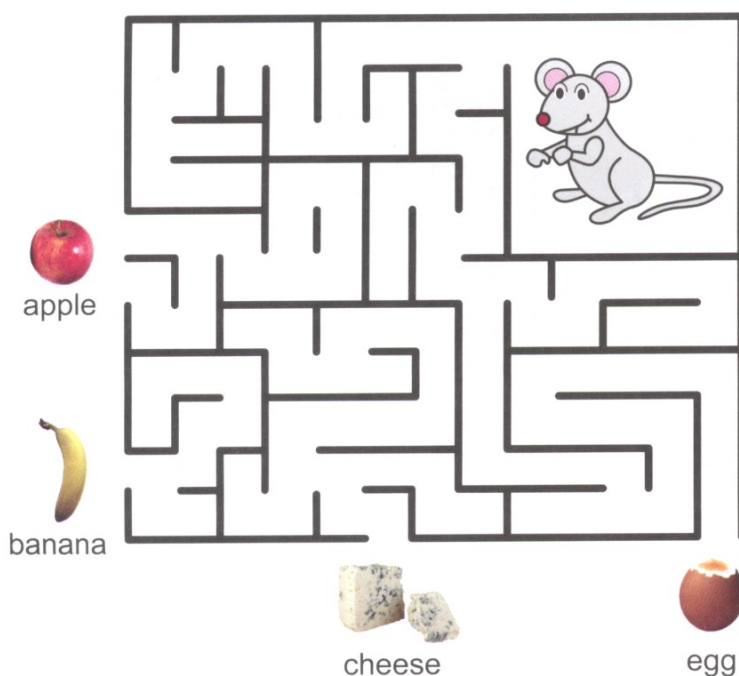
(C)



(D)



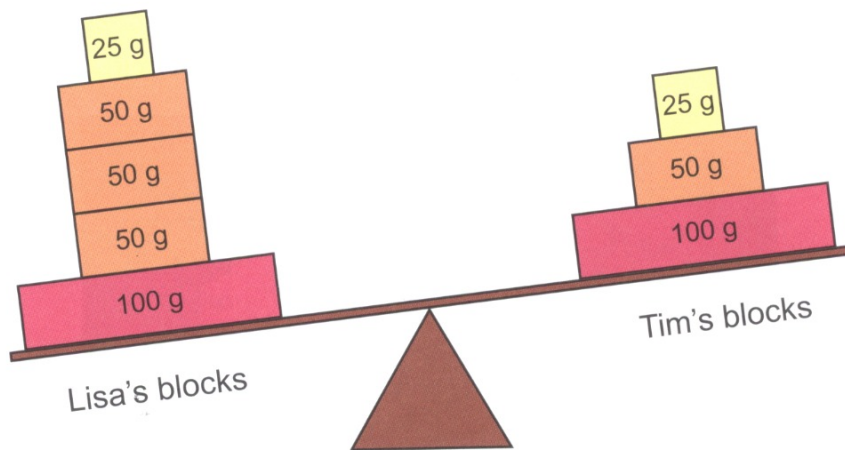
12. Mervyn the mouse is at the start of a maze. He can smell the food placed at the other end. The top of the maze is covered so the smell has to travel through the maze.



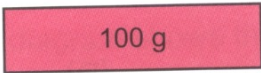
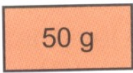
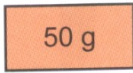
Which food can Mervyn smell?

- (A) apple
(B) banana
(C) cheese
(D) egg

13. Lisa and Tim were adding blocks to a seesaw.

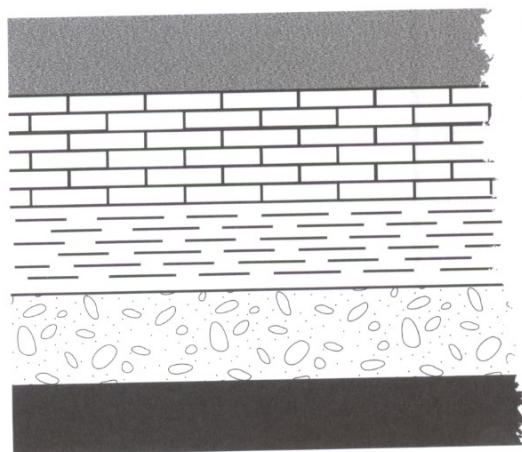


Which block(s) should Tim add to his side to balance the seesaw?

- (A)  + 
- (B)  + 
- (C)  + 
- (D) 

14. When rock layers form, new layers always form on top of the older ones.

This drawing shows some rock layers.



KEY

-  sandstone
-  limestone
-  shale
-  conglomerate
-  basalt

Which rock layer is the second youngest?

- (A) sandstone
(B) limestone
(C) shale
(D) conglomerate

For questions 15 and 16 use the information below.

Sasi wanted to make some green slime.

These are the instructions she used:

- 1) In one bowl, mix $\frac{1}{2}$ cup glue and $\frac{1}{2}$ cup water.
Add 3 drops food colouring.
- 2) In another bowl, mix 1 teaspoon borax with
1 cup water until the borax has dissolved.
- 3) Add the glue mixture to the borax solution,
stirring slowly.

15. Sasi already had a dropper, two bowls and a jug of water.
Her dad bought her borax, glue and food colouring.

What else did Sasi need to make her slime?

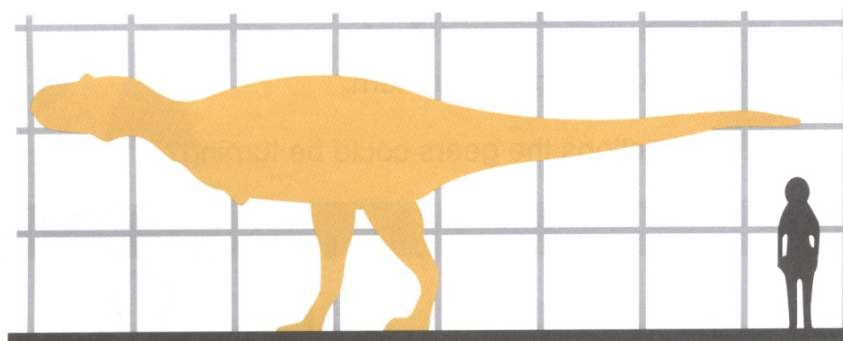
- (A) measuring cups and a teaspoon
- (B) a saucepan and a teaspoon
- (C) another bowl and a cup
- (D) a teaspoon and a mixer

16. How many cups of water did Sasi need altogether?

- | | |
|--------------------|-------|
| (A) $\frac{1}{2}$ | (B) 1 |
| (C) $1\frac{1}{2}$ | (D) 2 |



17. Amy drew a sketch of a person standing next to a Rajasaurus. She used square grid paper so that she could compare their sizes.



The person is one and a half metres tall.

How many metres long is the Rajasaurus from head to tail tip?

- | | | | |
|--------------------|-------|--------------------|--------------------|
| (A) $2\frac{1}{2}$ | (B) 7 | (C) $7\frac{1}{2}$ | (D) $8\frac{1}{2}$ |
|--------------------|-------|--------------------|--------------------|

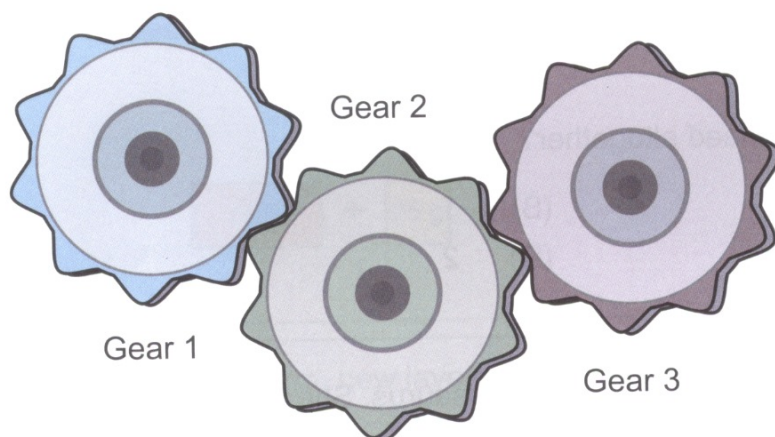
18. Priya got a pet mouse on her birthday, the 1st of January. It weighed 18 grams at the time. She then weighed the mouse at the end of each month for 4 months and recorded her results in this table.

At end of	How much the mouse weighed (grams)
January	21
February	26
March	30
April	33

During which month did the mouse grow the most?

- (A) January (B) February (C) March (D) April

19. The picture shows three gears.



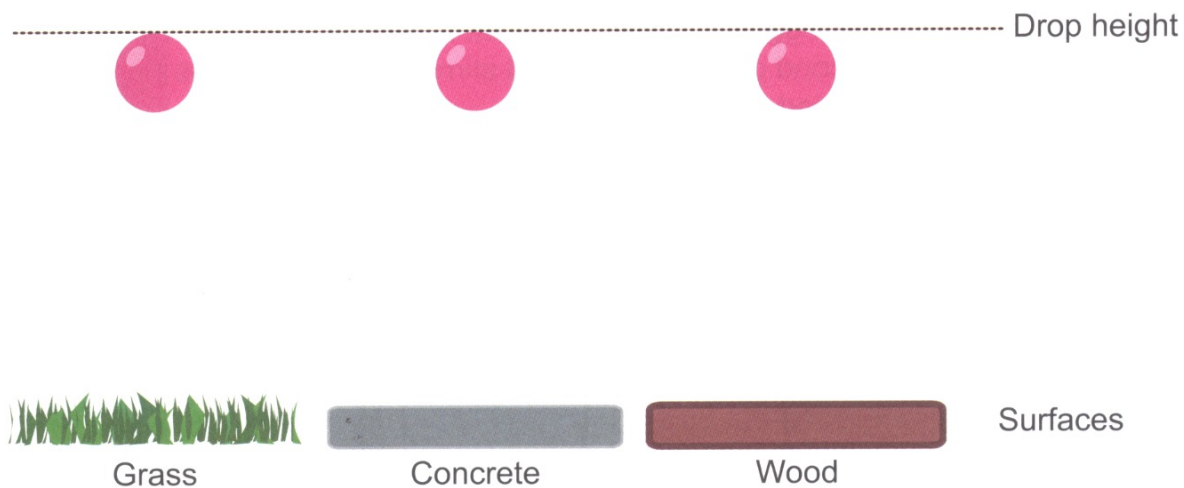
Each gear can turn this way ↻ or this way ↻.

Joe turns Gear 1 which makes the other gears turn.

Which of these shows the directions the gears could be turning?

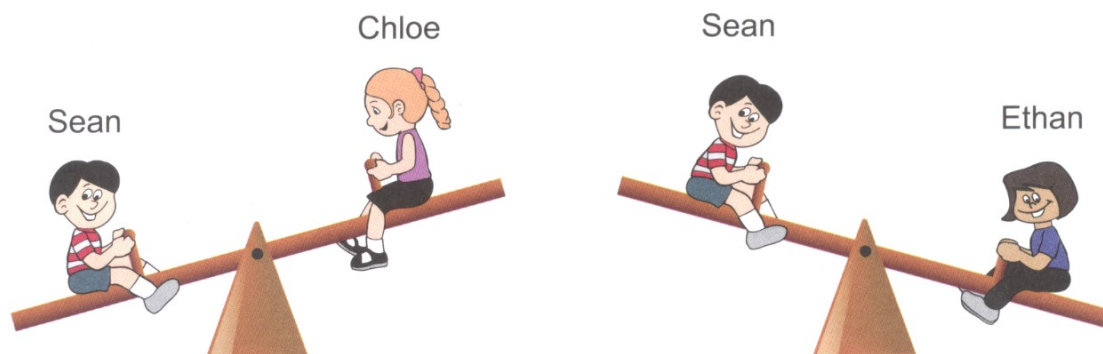
	Gear 1	Gear 2	Gear 3
(A)	↻	↻	↻
(B)	↻	↻	↻
(C)	↻	↻	↻
(D)	↻	↻	↻

20. Sam was testing the bounce of a rubber ball. The picture shows three of his tests.



What can Sam find out from these three tests?

- (A) whether the type of ball affects the bounce
 - (B) whether the type of surface affects the bounce
 - (C) whether the drop height affects the bounce
 - (D) whether the ball shape affects the bounce
21. Sean, Chloe, and Ethan are taking turns playing on a seesaw. Each time the seesaw stops, it becomes unbalanced, as shown.

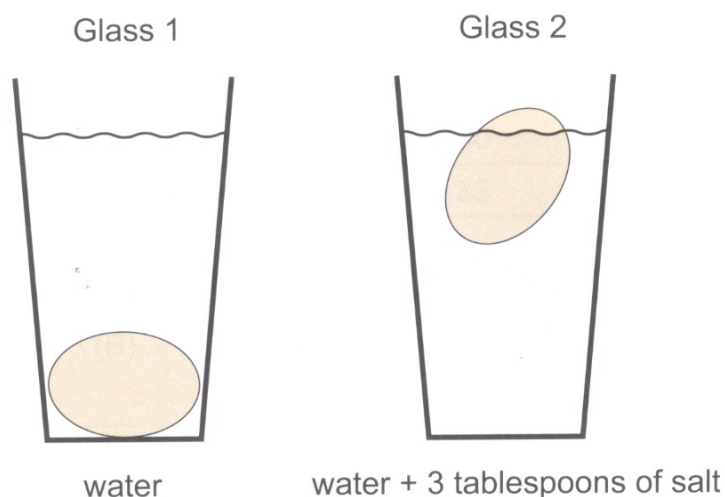


Who is the heaviest?

- (A) Chloe
- (B) Sean
- (C) Ethan
- (D) Chloe and Sean are the heaviest.

For questions 22 and 23 use the information below.

Mr Martin placed a fresh, raw egg into a glass of water. It sank to the bottom. He added 3 tablespoons of salt to a second glass of water and stirred it well. When he put the same egg into the salty water, it floated.



22. Mr Martin made another glass of salty water, using 1 tablespoon of salt.

Where would the egg settle in this glass of salty water?

- (A) the same as it did in Glass 1
(B) higher than it did in Glass 2
(C) the same as it did in Glass 2
(D) between where it did in Glass 1 and in Glass 2
23. Boiled eggs float better than fresh eggs. Mr Martin boiled this raw egg and tested it in the same glasses.

Which of these statements must be true?

- (A) It floated in the salty water. (B) It sank in the fresh water.
(C) It floated in the fresh water. (D) It sank in the salty water.

24. Four children wrote down the time it took them to get to school on a sunny day and on a rainy day.

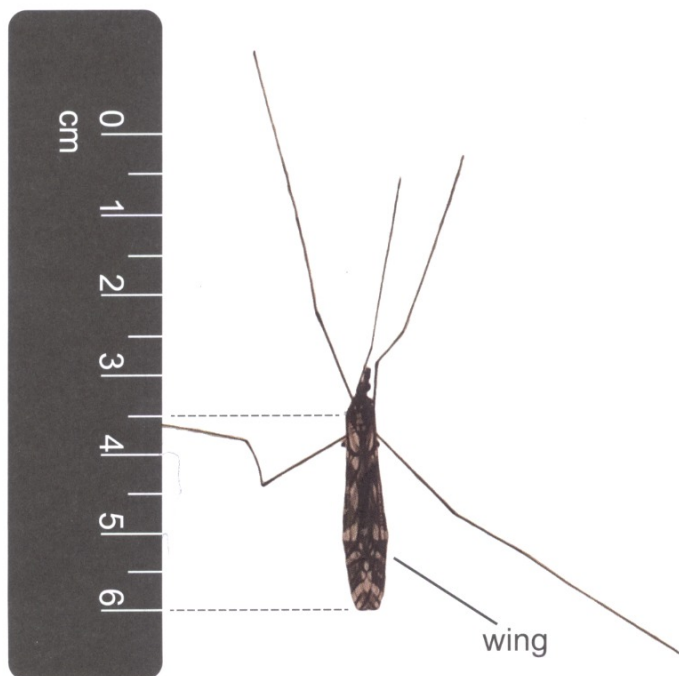
The table shows their times in minutes.

	Sunny day	Rainy day
Amy	10	15
Ben	15	20
Clara	5	15
David	20	20

Whose time changed the most from a sunny day to a rainy day?

- (A) Amy (B) Ben (C) Clara (D) David

25. Xiang found a large mosquito on the wall. She placed a ruler next to it and took a photo.



How long is the wing of the mosquito in centimetres (cm)?

- (A) $2\frac{1}{2}$ (B) $3\frac{1}{2}$ (C) $4\frac{1}{2}$ (D) 6

26. Milan has three bags from different supermarkets.



He puts a weight into a bag and lifts it. He puts more weights in until the bag breaks and records the number of weights he used.

He repeats this with the other two bags.

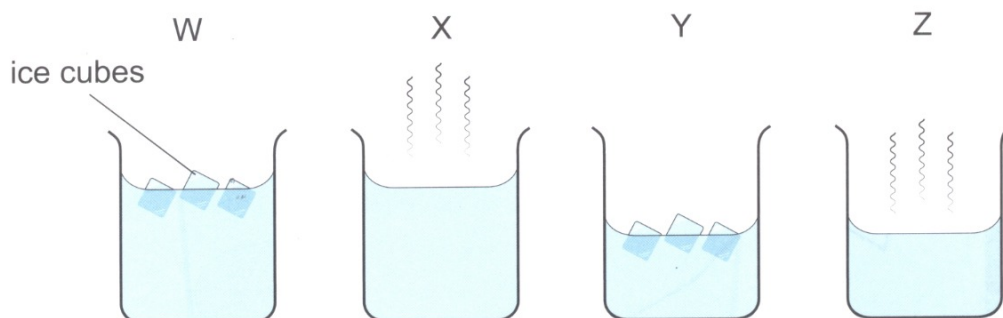
What question can Milan answer using his results?

- (A) Which bag handle is the most comfortable for carrying the weights?
(B) Which bag looks best carrying the weights?
(C) Which bag best fits the shape of weights?
(D) Which bag can carry the most weights?

27. Adam knows that he can dissolve more sugar in hot water than he can in an equal amount of cold water.

He also knows that, at any temperature, the more water he has, the more sugar he can dissolve in it.

Adam has four cups of water.



He wants to order the cups according to how much sugar he can dissolve in each one.

Which two cups will be hardest to place in the correct order?

- (A) cup W and cup Z (B) cup X and cup Z
(C) cup W and cup Y (D) cup X and cup Y

For questions 28 and 29 use the information below.

The table shows the components that make up 100 grams of cow's milk.

Component	water	milk sugar	fat	protein	minerals
Amount (grams)	87	5	4	3	1

Only water is a liquid at room temperature. The rest of the components are solids.

28. How many grams of solids that are **NOT** fat are in 100 grams of cow's milk?

- (A) 13 (B) 9 (C) 6 (D) 4

29. Milk powder is made by removing the water from the milk.

How many grams of milk sugar is in milk powder made from 1000 grams of cow's milk?

- (A) 500 (B) 130 (C) 90 (D) 50

30. David has four rocks.

Rock 1



Rock 2



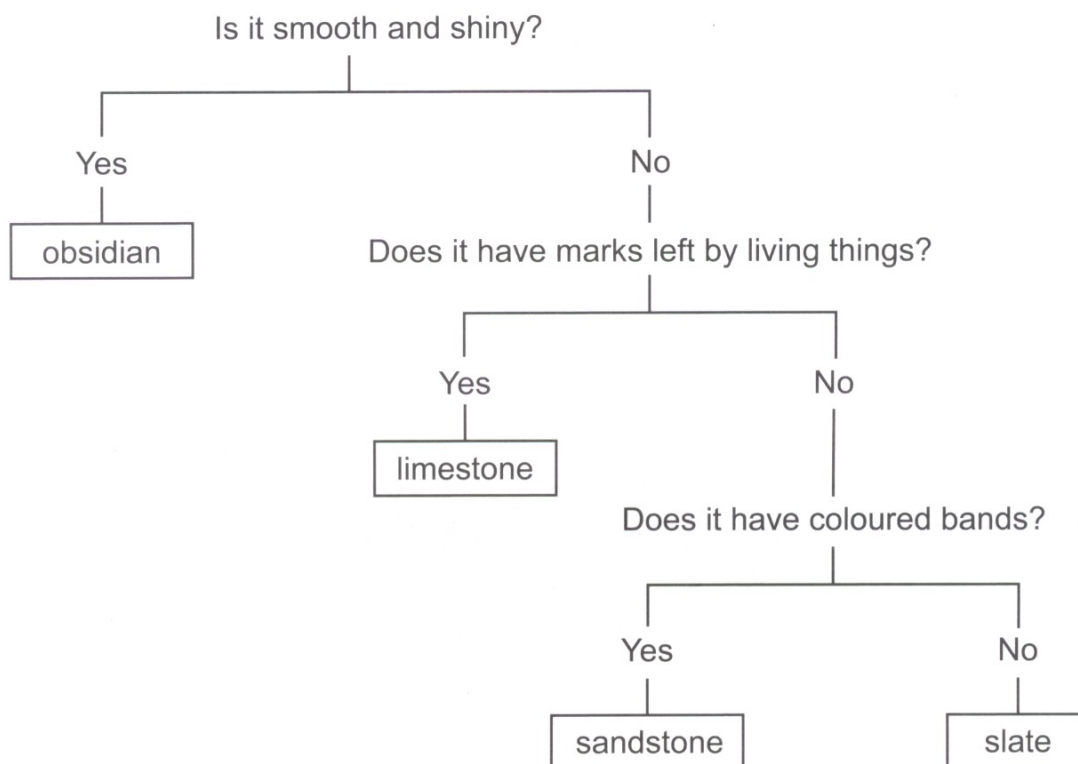
Rock 3



Rock 4



He came up with a key so that his friends could work out the name of each of his rocks.



Which of the following gives the correct name of each rock?

	Rock 1	Rock 2	Rock 3	Rock 4
(A)	obsidian	limestone	sandstone	slate
(B)	slate	sandstone	obsidian	limestone
(C)	obsidian	slate	limestone	sandstone
(D)	sandstone	slate	obsidian	limestone